Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/16/2005
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Foot Reaction Forces During Space Flight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cavanagh@u.washington.edu	**Fax:**	FY 206-685-3139
PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	07/01/1998	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	ISS		
Flight Assignment:	ISS Expeditions 6, 8, 11, 12 In flight development phase (data collection has begun) NOTE: End date is now 4/30/2008 (chgd from 4/30/2007) per PI (5/07)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NCC9-153		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Lower-extremity loading during daily life in microgravity is thought to be markedly reduced compared with loading in 1g, although no measurements have yet been made to quantify the load reduction. Recent studies have shown that as much as 1.58% per month of bone mineral is lost from the proximal femur during 4- to 14-month flights and that greater than 20% of knee-extensor strength is lost in 60- to 80-day flights. In this study, we propose to quantify mechanical load during periods of time which represent typical activities of daily life, both on earth and during a mission on the International Space Station (ISS). Hardware designed for ambulatory data collection on the Human Research Facility (HRF) will be used to collect ground reaction force, EMG, and three lower-extremity joint angles for four periods on Earth and four periods on the ISS of approximately 8 hours on each of the crew members while they continue with their other assigned tasks. Daily mechanical load stimulus will be calculated using two competing theories, and daily neural drive will be estimated for the measurements obtained. DXA scans of the whole body, MRI of the leg and arm muscles and maximum ankle, knee, elbow, and hip joint torques will be taken pre- and post-flight to quantify any bone density, muscle volume, and muscle strength changes which have occurred during the flight period. The results will allow a direct comparison of two daily load-stimulus estimates with the directly measured changes in BMD and with neural drive between 1g and 0g environments. The prevalence of eccentric and concentric muscle actions under the two gravitational conditions will also be determined. This project will lay an important foundation for the continuing development of countermeasures to lower-extremity muscle and bone loss on future space station missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this research is to quantify lower-extremity loading, muscle activation, and joint range of motion during exercise in both 0g and 1g. This information will provide further insight into the role of skeletal unloading in bone loss during long-term space flight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knoweldge in ths area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	In the last 12 months, all technical and administrative goals have been met. All data for the second subject (Expedition 8) have been collected, including in-flight sessions, DXA, MRI, and strength testing, and an Operational Accomplishments Report has been submitted. In addition, baseline ambulatory data was collected in 1g for all remaining subjects. Unfortunately, two subjects have been withdrawn from the experiment due to crew swaps and manifest space limitations. Ambulatory data collections for all currently enrolled subjects should be complete in early FY2005. Materials have been purchased to manufacture a customized Lower Extremity Monitoring Suit (LEMS) should additional subjects need to be recruited. Customized data analysis routines in MATLAB have been enhanced and data analysis is in progress. EMG analysis is being performed in collaboration with a team at Zin Technologies (Cleveland, OH). Data from this experiment were presented at the 2004 Annual ASBMR meeting held in Seattle, WA and at the 2004 Research ShowCASE held at Case Western Reserve University in Cleveland, OH. Several manuscripts are in preparation for submission in FY2005. This data was also presented at Grand Rounds at Johnson Space Center and to the Space Station Program Control Board (SSPCB).
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Rice AJ, Maender CC, Genc KO, Ochia RS, Snedeker JG, Cavanagh PR. "Bone Loss and Lower Extremity Loading During Long-Duration Space Flight." Twenty-Sixth Annual Meeting of the American Society for Bone and Mineral Research, October 2004. J Bone Miner Res, 2004 Oct;19 Suppl:S93. , Oct-2004
Abstracts for Journals and Proceedings	Genc KO, Maender CC, Rice AJ, Ochia RS, Snedeker JG, Cavanagh PR. "Treadmill Exercise on the International Space Station: The Effects of External Loading" The Twenty-Sixth Annual Meeting of the American Society for Bone and Mineral Research, October 2004. J Bone Miner Res, 2004 Oct;19 Suppl:S452. , Oct-2004
Articles in Peer-reviewed Journals	Cavanagh PR, Licata AA, Rice AJ. "Exercise and pharmacological countermeasures for bone loss during long-duration" Gravit Space Biol Bull. 2005 Jun;18(2):39-58. Review. PMID: 16038092 , Jun-2005
Articles in Peer-reviewed Journals	Cavanagh PR, Genc KO, Gopalakrishnan R, Kuklis MM, Maender CC, Rice AJ. "Foot forces during typical days on the international space station." J Biomech. 2010 Aug 10;43(11):2182-8. PMID: 20462584 [note added by Task Book editor--10/5/10] , Aug-2010
Articles in Peer-reviewed Journals	Genc KO, Gopalakrishnan R, Kuklis MM, Maender CC, Rice AJ, Bowersox KD, Cavanagh PR. "Foot forces during exercise on the International Space Station." J Biomech. 2010 Aug 19. [Epub ahead of print] PMID: 20728086 [note added by Task Book editor--10/5/10] , Aug-2010
Significant Media Coverage	NBC Medical Miracles. "New Frontiers in Research: Bone Density, Stem Cells, Macular Degeneration." NBC Medical Miracles, April 2004., Apr-2004